Week 10 Progress Report

Details

- **Name of Student:** Tirthak Girish Likhar
- **PRN:** 22070521041
- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (09/03/2026 to 13/03/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of a Week

Week 10 marked the resumption of hands-on lab work following the Nullcon Goa conference attended during Weeks 8 and 9. The primary focus this week was on rebooting the Wazuh SIEM environment, resolving a dynamic IP change on the Ubuntu Server VM, reconnecting the Windows 11 Wazuh agent, and executing the first real attack simulations in the lab.

Upon booting the Ubuntu Server VM, the Wazuh Manager, Indexer, and Dashboard services were verified as active. However, the server had received a new dynamic IP address (192.168.1.11) after reboot — previously it was 192.168.1.10. The Windows agent's ossec.conf was updated to reflect the new IP, the Wazuh service was restarted on the Windows endpoint, and the agent was confirmed as Active in the Wazuh Dashboard with 100% agent coverage.

With the environment stable, the first attack simulation of the project was performed: a failed Windows login brute-force scenario. Six consecutive incorrect password attempts were made on the Windows 11 endpoint (Windows Endpoint Agent – Windows 11 monitored endpoint). Wazuh detected all six events in real-time, generating Event ID 4625 (Logon Failure) alerts with Rule ID 60122 at severity Level 5, mapping to MITRE ATT&CK technique T1078 (Valid Accounts). The alerts were investigated in the Wazuh Security Events dashboard, confirming full end-to-end SIEM detection capability.

Objectives of the Week

- **To reboot the Wazuh SIEM environment and verify all three core services (Manager, Indexer, Dashboard) were active after a multi-week pause.**
- **To resolve the Ubuntu Server's dynamic IP change and restore communication between the Windows Wazuh agent and the SIEM manager.**
- **To perform the first real attack simulation in the lab — a failed Windows login brute-force — and verify Wazuh detection in real-time.**
- **To investigate the generated alerts using the Wazuh Security Events dashboard, including Event ID, Rule ID, MITRE ATT&CK mapping, and compliance tags.**
- **To document the dynamic IP issue and plan for static IP assignment on the Ubuntu Server VM in an upcoming session.**

Key Accomplishments

- Rebooted and verified the complete Wazuh SIEM stack — Manager, Indexer, and Dashboard all confirmed active using **systemctl is-active**.
- Identified and resolved a dynamic IP change on the Ubuntu Server VM (192.168.1.10 → 192.168.1.11) by updating the **<address>** field in the Windows agent's ossec.conf.
- Restarted the Windows Wazuh Agent service (wazuhsvc) and confirmed Windows Endpoint Agent showing as **Active** in the Wazuh Dashboard.
- **Executed Attack Simulation 1: Failed Windows Login Brute-Force** — six consecutive incorrect password attempts on the Windows 11 endpoint.

- All six attempts were detected by Wazuh, generating **Event ID 4625** (Logon Failure) alerts with **Rule ID 60122** at severity Level 5.
- Performed full alert investigation in the Wazuh Security Events dashboard — verified MITRE ATT&CK mapping to **T1078 (Valid Accounts)**, compliance tags (GDPR, HIPAA, PCI-DSS, NIST 800-53), and timeline spike visualization.
- Confirmed Logon Type 2 (Interactive), failure reason (Unknown user name or bad password), source process (svchost.exe), and source IP (127.0.0.1) from the alert metadata.

Issues Faced & Resolutions – Week 10

- **1. Ubuntu Server dynamic IP change after reboot**

  Issue: After rebooting the Ubuntu Server VM, the IP address changed from 192.168.1.10 to 192.168.1.11. The Windows Wazuh agent's ossec.conf still pointed to the old IP, causing the agent to show as Disconnected in the dashboard.

  Resolution: Used 'ip a' on the Ubuntu Server to identify the new IP address. Updated the <address> field in C:\Program Files (x86)\ossec-agent\ossec.conf to 192.168.1.11 and restarted the wazuhsvc service via PowerShell. Agent confirmed Active in the dashboard. Static IP assignment planned for a future session.

- **2. Ensuring all Wazuh services active after multi-week pause**

  Issue: After Weeks 8 and 9 (Nullcon attendance), the Wazuh server had been shut down for an extended period. Services needed to be verified before any lab work could begin.

  Resolution: Ran 'sudo systemctl is-active wazuh-manager', 'wazuh-indexer', and 'wazuh-dashboard' sequentially. All three returned 'active'. Proceeded with lab work after confirmation.

Tools Used – Week 10

- **Wazuh SIEM (Manager + Indexer + Dashboard v4.7.5)** – Primary detection and monitoring platform; used for alert generation, rule evaluation, and investigation.
- **VirtualBox** – Hypervisor hosting the Ubuntu Server 24.04 VM (Wazuh server) and Ubuntu 22.04 attacker VM.
- **Ubuntu Server 24.04 (VM)** – Hosts the Wazuh SIEM stack; SSH accessed from Windows host for service management.
- **Wazuh Agent 4.7.5 (Windows 11 Endpoint)** – Installed on host machine; forwards Windows Security Event logs to the SIEM.
- **Windows PowerShell** – Used to restart wazuhsvc and verify agent service status.
- **Windows Event Log (Event Viewer)** – Source of Event ID 4625 (Logon Failure) logs forwarded to Wazuh.

Conclusion of Week 10

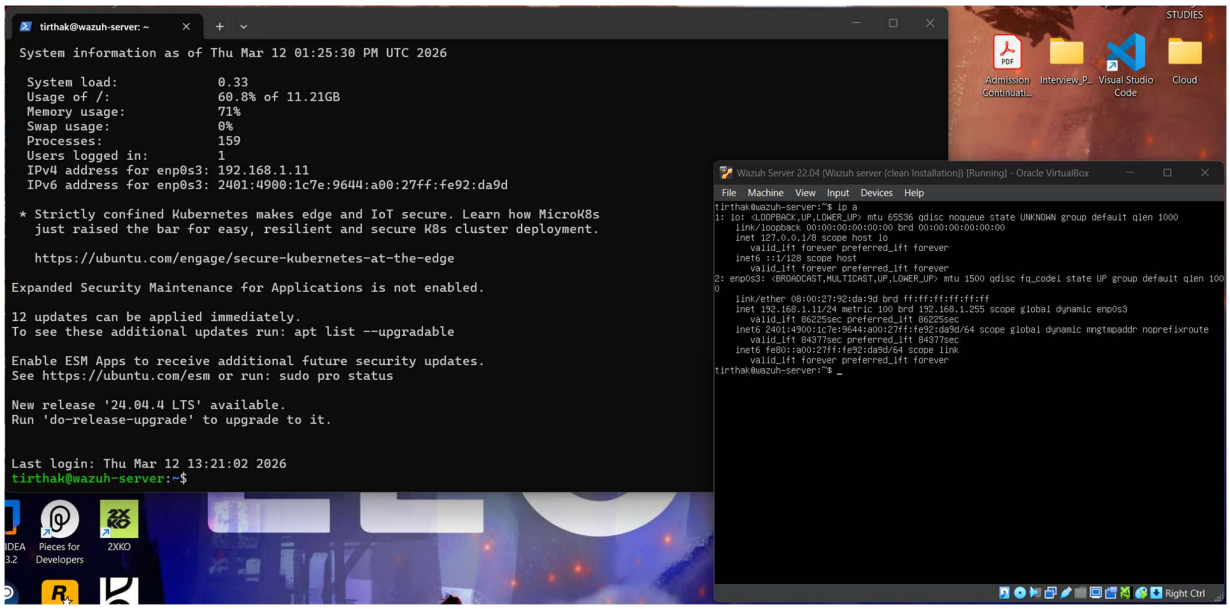
Week 10 marked a significant transition from conference attendance back to active hands-on lab work. The Wazuh SIEM environment was successfully restored after a multi-week pause, with all three core services verified as running and the Windows agent reconnected after resolving a dynamic IP issue on the Ubuntu Server VM.

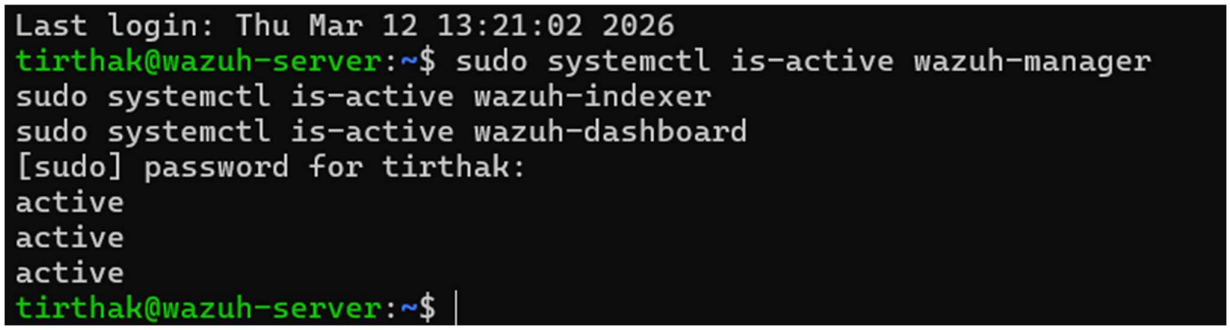
The first real attack simulation of the internship project — a failed Windows login brute-force — was successfully executed and detected by Wazuh in real-time. The complete alert investigation cycle was completed: from generating the attack, observing the Event ID 4625 spike in the Security Events dashboard, to mapping the detection to MITRE ATT&CK technique T1078 and verifying compliance framework tags. This represents the first practical demonstration of the SIEM pipeline performing its core function of threat detection and alerting.

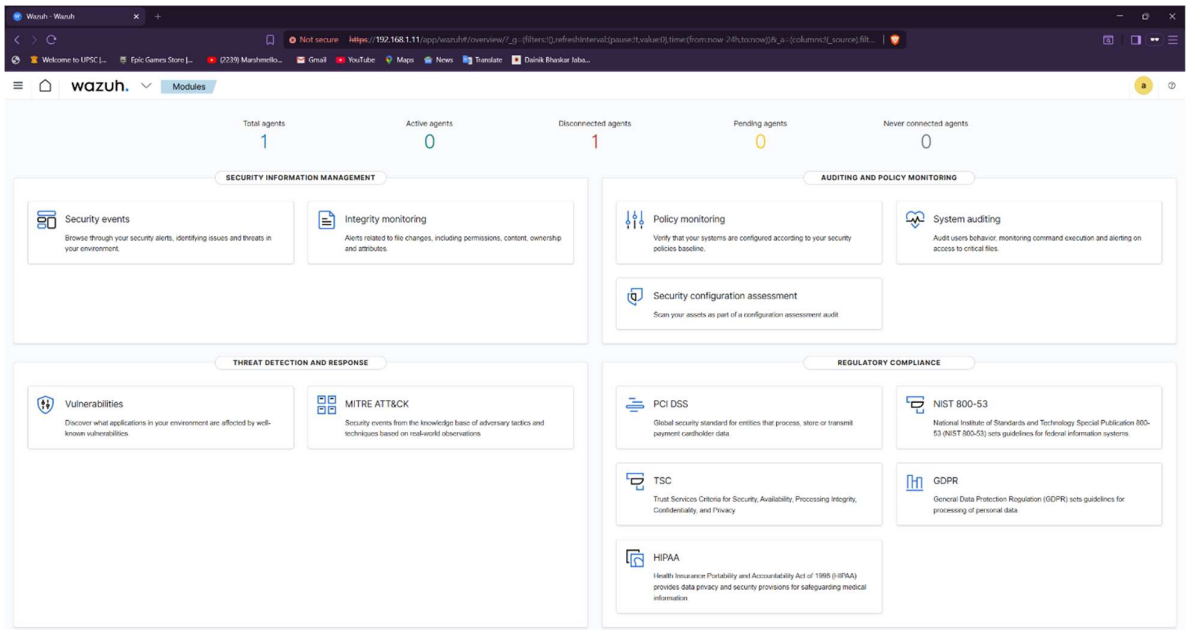
The dynamic IP issue encountered this week also reinforced the need to configure a static IP on the Ubuntu Server VM, which will be addressed in Week 11 alongside further attack simulations including File Integrity Monitoring (FIM) triggers and SSH brute-force from the Ubuntu attacker machine.

Screenshots of Activities (if Any)

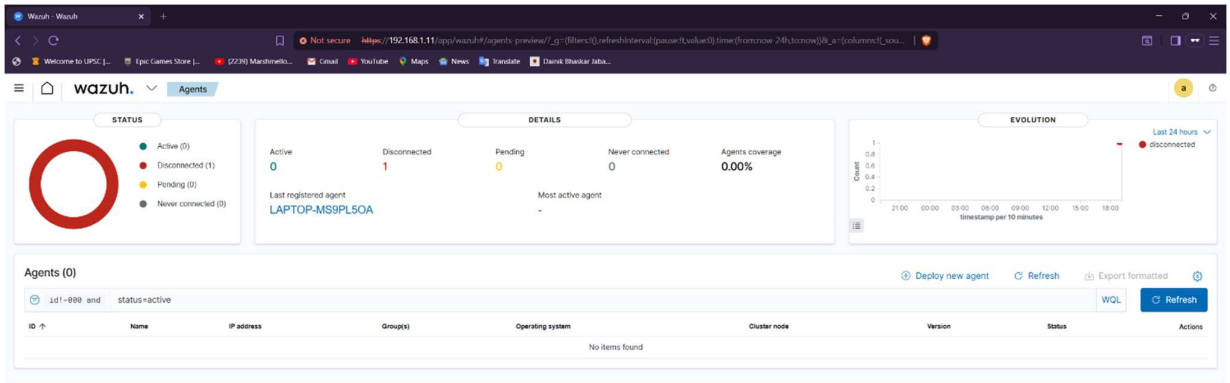
[Please insert screenshots from your Wazuh dashboard here — Wazuh services active, agent active, Event ID 4625 alerts, MITRE mapping, timeline spike]





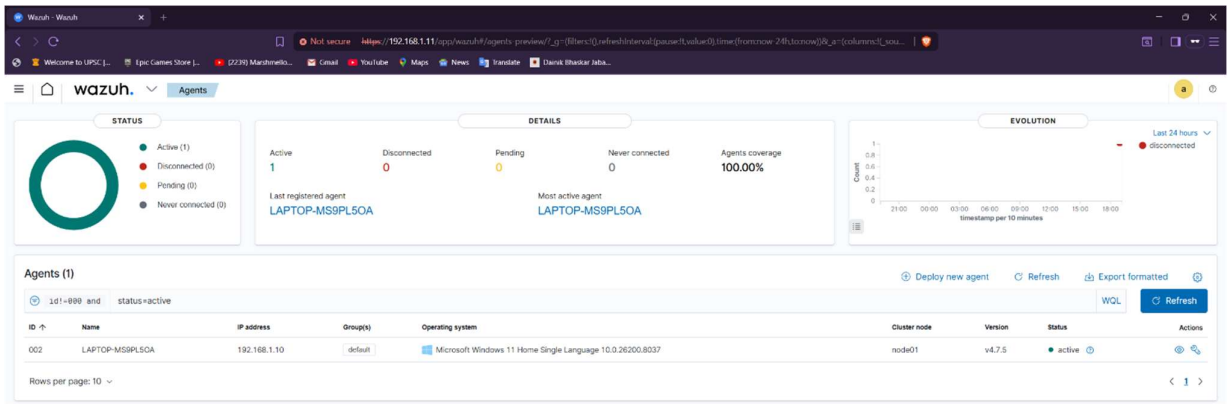


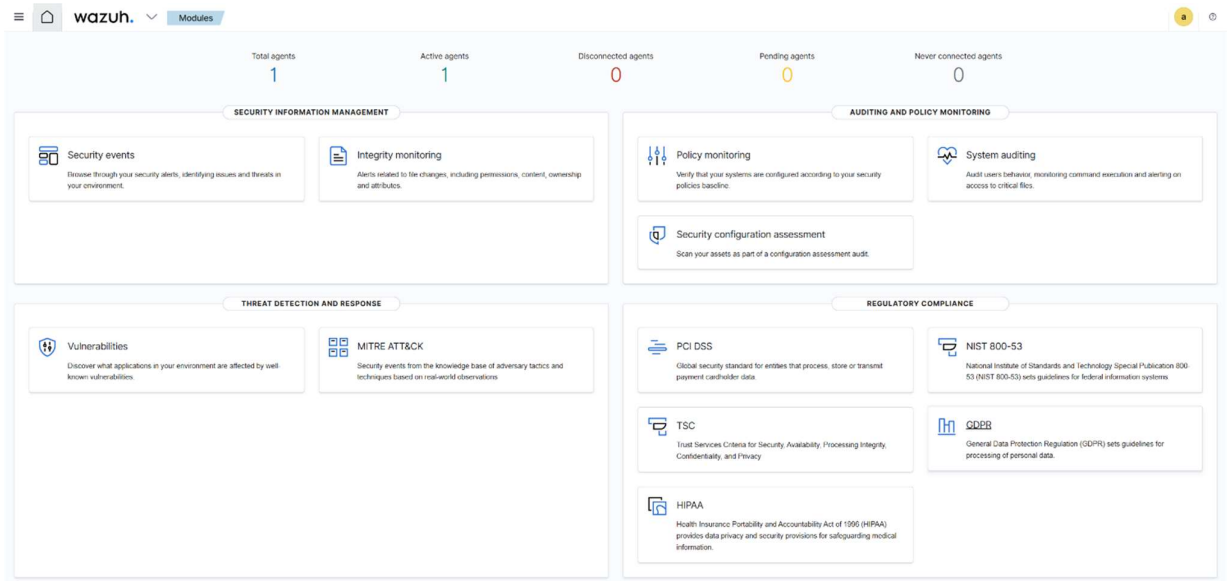
```
Administrator: Windows Powe

PS C:\Users\Tirthak> Start-Service wazuhsvc
PS C:\Users\Tirthak> Get-Service wazuhsvc

Status   Name                DisplayName
------   ----                -----------
Running  wazuhsvc            Wazuh


PS C:\Users\Tirthak>
```

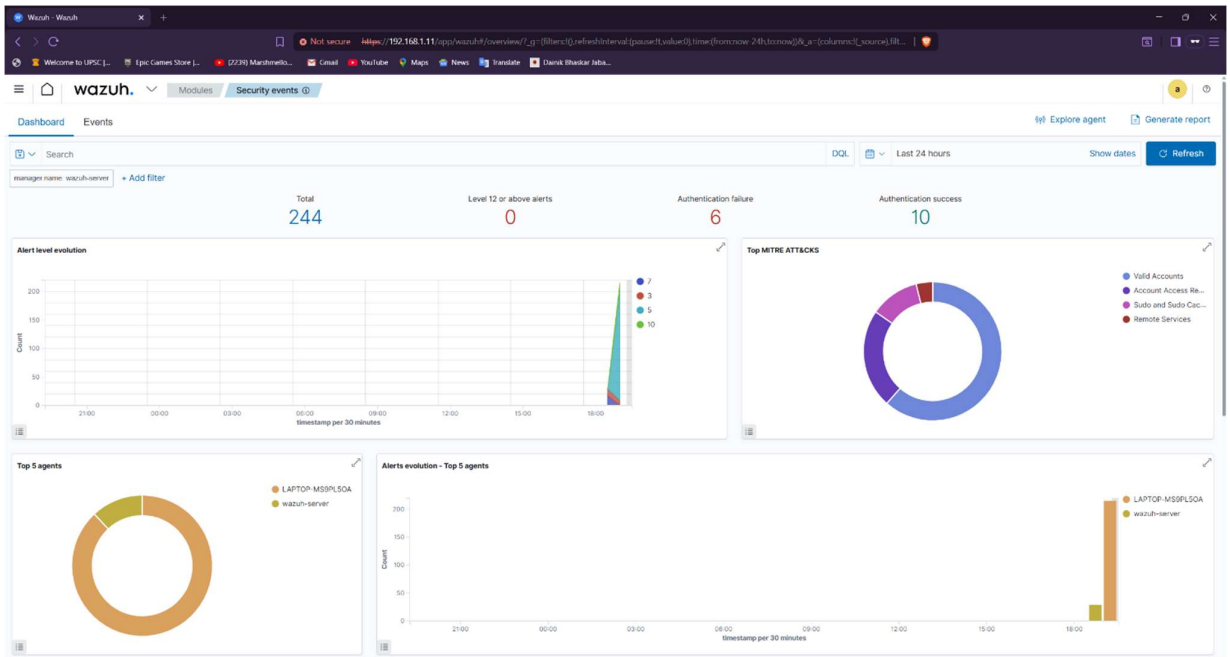


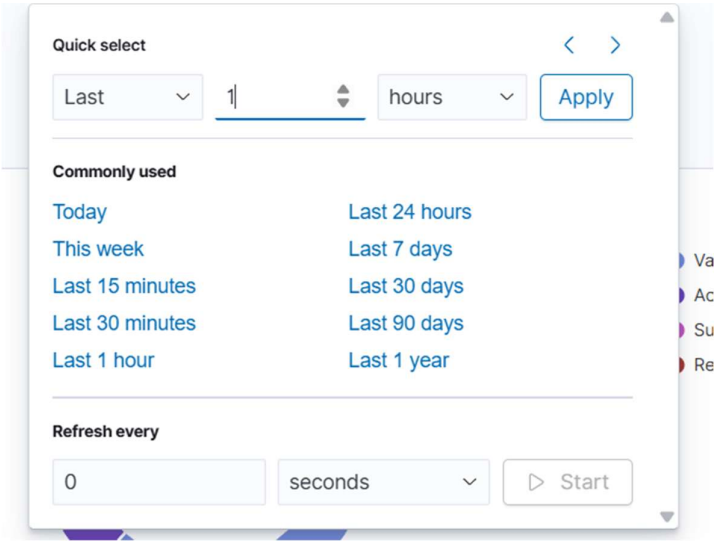

```
<!--
  Wazuh - Agent - Default configuration for Windows
  More info at: https://documentation.wazuh.com
  Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>

  <client>
    <server>
      <address>192.168.1.10</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>windows, windows10</config-profile>
    <crypto_method>aes</crypto_method>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
  </client>

  <!-- Agent buffer options -->
  <client_buffer>
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Log analysis -->
  <localfile>
    <location>Application</location>
    <log_format>eventchannel</log_format>
  </localfile>

  <localfile>
```

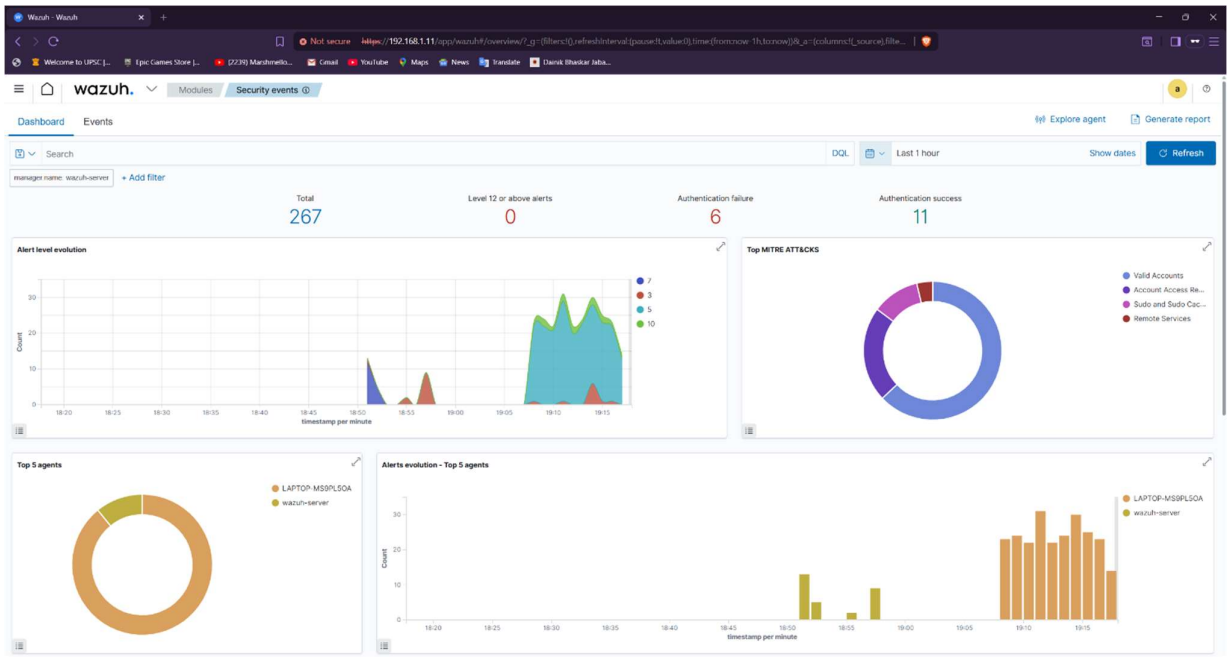
```
PS C:\Users\Tirthak> Stop-Service wazuhsvc -Force
PS C:\Users\Tirthak> Start-Service wazuhsvc
PS C:\Users\Tirthak> Get-Service wazuhsvc

Status   Name                DisplayName
------   ----                -----------
Running  wazuhsvc            Wazuh


PS C:\Users\Tirthak>
```

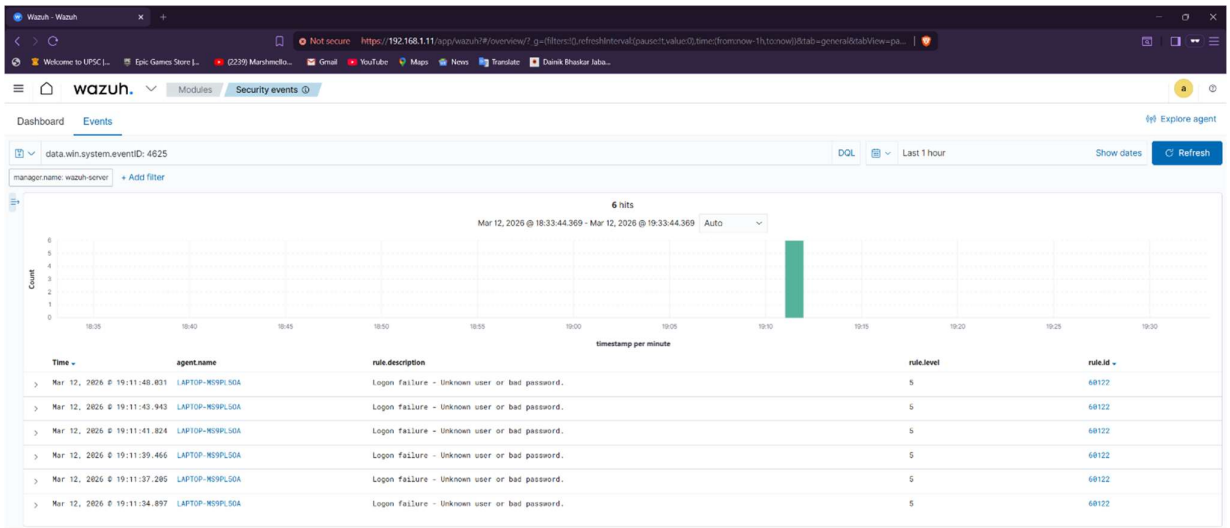





Wazuh - Wazuh
Not secure https://192.168.1.11/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&_a=(columns:!(_source),filt...
wazuh.
Modules
Security events
Dashboard
Events
Explore agent
Generate report
Search
DQL
Last 24 hours
Show dates
Refresh
manager.name: wazuh-server
+ Add filter
Total
244
Level 12 or above alerts
0
Authentication failure
6
Authentication success
10
Alert level evolution
Count
timestamp per 30 minutes
Top MITRE ATT&CKS
Valid Accounts
Account Access Re...
Sudo and Sudo Cac...
Remote Services
Top 5 agents
LAPTOP-MS9PL5OA
wazuh-server
Alerts evolution - Top 5 agents


Quick select
Last
1
hours
Apply
Commonly used
Today
This week
Last 15 minutes
Last 30 minutes
Last 1 hour
Last 24 hours
Last 7 days
Last 30 days
Last 90 days
Last 1 year
Refresh every
0
seconds
Start


Wazuh - Wazuh
Not secure https://192.168.1.11/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-1h,to:now))&_a=(columns:!(_source),filte...
wazuh.
Modules
Security events
Dashboard
Events
Explore agent
Generate report
Search
DQL
Last 1 hour
Show dates
Refresh
manager.name: wazuh-server
+ Add filter
Total
267
Level 12 or above alerts
0
Authentication failure
6
Authentication success
11
Alert level evolution
Count
timestamp per minute
Top MITRE ATT&CKS
Valid Accounts
Account Access Re...
Sudo and Sudo Cac...
Remote Services
Top 5 agents
LAPTOP-MS9PL5OA
wazuh-server
Alerts evolution - Top 5 agents


Wazuh - Wazuh
Not secure https://192.168.1.11/app/wazuh?#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-1h,to:now))&tab=general&tabView=pa...
Welcome to UPSC |...
Epic Games Store |...
(2239) Marshmello...
Gmail
YouTube
Maps
News
Translate
Dainik Bhaskar Jaba...
wazuh.
Modules
Security events
Dashboard
Events
Explore agent
data.win.system.eventID: 4625
DQL
Last 1 hour
Show dates
Refresh
manager.name: wazuh-server
+ Add filter
6 hits
Mar 12, 2026 @ 18:33:44.369 - Mar 12, 2026 @ 19:33:44.369
Auto
Count
timestamp per minute
Time
agent.name
rule.description
rule.level
rule.id
Mar 12, 2026 @ 19:11:48.031 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122
Mar 12, 2026 @ 19:11:43.943 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122
Mar 12, 2026 @ 19:11:41.824 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122
Mar 12, 2026 @ 19:11:39.466 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122
Mar 12, 2026 @ 19:11:37.205 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122
Mar 12, 2026 @ 19:11:34.897 LAPTOP-MS9PLSOA Logon failure - Unknown user or bad password. 5 60122